

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points.

**Problem 1**

Find the absolute maximum and minimum values of  $f(x) = -\frac{1}{x^2}$  on  $[0.5, 2]$ . Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

**Problem 2**

Suppose that  $f'(x) = 2x$  for all  $x$ . Find  $f(3)$  given

a)  $f(0) = 2$

b)  $f(1) = 0$

c)  $f(0) = 4$

**Problem 3**

Let  $h(x) = -x^3 + 2x^2$ . Find (a) the open intervals on which the function is increasing and decreasing, and (b) identify the function's local and absolute extreme values, if any, saying where they occur.

**Problem 4**

Identify the inflection points and local maxima and minima of  $y = (x - 2)^3 + 1$ . Identify the intervals on which the function is concave up and concave down.

**Problem 5**

You have been asked to design a one-liter ( $1000 \text{ cm}^3$ ) can shaped like a right circular cylinder. What dimensions will use the least material? Hint: your trying to minimize the surface area of the can.

**Problem 6**

Use Newton's method to get a decimal approximation of  $\sqrt{7}$  from  $f(x) = x^2 - 7$ . Start with  $x_0 = 2$  and find  $x_3$ . Then calculate  $f(x_3)$ .

**Problem 7**

Find an antiderivative for each function (you only need to find a single antiderivative, not a family of functions).

a)  $f(x) = x^{-4} + 2x + 3$

b)  $f(x) = \frac{5}{x^2}$

c)  $f(x) = 3\sin(2x)$

**Problem 8**

Solve:  $\int(\sqrt{x} + 2x^3)dx$ .

**Extra Credit:**

The standard equation for the position  $s$  of a body moving with a constant acceleration  $a$  along a coordinate line is  $s(t) = s_0 + v_0t + \frac{1}{2}at^2$  where  $v_0$  and  $s_0$  are the body's velocity and position at time  $t = 0$ . Derive the equation for  $s(t)$  from the fact that  $\frac{d^2s}{dt^2} = a$  where  $a$  is a constant. It is given that  $\frac{ds}{dt} = v_0$  at  $t = 0$ , and that  $s(t) = s_0$  at  $t = 0$ .